EXPLAINABLE ViT-B/32 FOR ALZHEIMER'S MRI CLASSIFICATION

Journal Submission (Elsevier Neuroscience Informatics NEURI-D-26-00260R1) & Empirical Benchmarks

DOC ID: RT-RES-NEURI-2026-V1
Principal Architect: Bassey Riman
Journal: Neuroscience Informatics | alzdetect.live
STATUS: VERIFIED // PAGE 1 OF 3

EXECUTIVE RESEARCH & PUBLICATION SUMMARY

This dossier documents the peer-reviewed scientific research led by Bassey Riman (First & Corresponding Author, in collaboration with co-authors Suraj Verma, Claudio Angione, and Annalisa Occhipinti of Teesside University, U mitted for publication in Elsevier's Neuroscience Informatics (Manuscript Ref: NEURI-D-26-00260R1). The 20-page study establishes an integrated experimental framework for 4-stage Alzheimer's d MRI classification (NonDemented, VeryMildDemented, MildDemented, ModerateDemented), combining two-stage ViT-B/32 domain adaptation, controlled 3-seed reproducibility, backbone unfreezing ablations, quantitative perturbation-based attention faithfulness (deletion/insertion AUC), and translational deployment via https://www.alzdetect.live/.

JOURNAL SUBMISSION & BENCHMARK METRICS

Target Journal:	Neuroscience Informatics (Elsevier)
Manuscript Ref:	NEURI-D-26-00260R1 (20 Pages)
Lead Author:	Bassey Riman* (Corresponding Author)
Portal Status:	Submitted / Under Review (Oct 2026)
Primary Test Acc:	98.84% (n=6,400) | Macro F1: 99.24%
Faithfulness Adv:	Del +0.12455 | Ins +0.15387 (p<0.05)
Live Platform:	https://www.alzdetect.live/

01 PEER-REVIEWED MANUSCRIPT & EDITORIAL MANAGER SUBMISSION PROOF

Elsevier NEURI-D-26-00260R1

MANUSCRIPT PAGE 1: TITLE, AUTHORS & ABSTRACT

An Explainable Vision Transformer Framework for Stage-Wise Alzheimer's MRI Image Classification and an Interactive Research Prototype

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ARTICLE INFO

Keywords:
Vision Transformer (ViT)
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Magnetic Resonance Imaging (MRI)
Explainable Artificial Intelligence (XAI)
Attention Faithfulness
Backbone Fine-Tuning
Research Prototype

ABSTRACT

Purpose: Early-stage Alzheimer's disease (AD) classification from structural magnetic resonance imaging (MRI) remains challenging because anatomical changes may be subtle and spatially distributed, while deep-learning predictions can be difficult to interpret. This study presents an integrated framework combining stage-wise MRI image classification, comparative modelling, explainability evaluation, and interactive research deployment.

Methods: A pre-trained Vision Transformer (ViT-B/32) was adapted for four image-level categories: NonDemented, VeryMildDemented, MildDemented, and ModerateDemented, using a derived ADNI-based image dataset. The framework was evaluated against baseline ViT and CNN implementations. Additional experiments assessed reproducibility across three random seeds, frozen and partial backbone adaptation, and quantitative attention faithfulness through patch deletion and insertion.

Results: The original fine-tuned ViT evaluation achieved 98.84% accuracy on a 6,400-image held-out test partition. Across three additional full-fine-tuning runs, mean test accuracy was 98.47% ± 0.53 percentage points. A seed-42 backbone ablation yielded accuracies of 55.11%, 80.44%, and 98.11% for frozen, partially fine-tuned, and fully fine-tuned configurations, respectively. On 80 correctly classified test images, attention-guided patch ordering outperformed random ordering, with mean deletion and insertion AUC advantages of 0.12455 and 0.15387, respectively. The classification and visual explanation workflow was integrated into AlzDetect, an interactive research prototype.

Conclusions: These findings demonstrate promising within-dataset performance and the feasibility of integrating classification, explanation analysis, and software deployment. However, image-level partitioning, possible subject overlap, unmatched comparison conditions, and limited explanation evaluation constrain generalisability. Subject-independent and external multi-centre validation are required before clinical utility can be established.

Ref: Riman et al. (2026) | 20 Pages, 8 Figures, 4 Tables

OFFICIAL ELSEVIER EDITORIAL MANAGER SUBMISSION RECORD (NEURI-D-26-00260R1)



Verified Author Portal (editorialmanager.com/neuri): Manuscript Number NEURI-D-26-00260R1 | Title: 'An Explainable Vision Transformer Framework for Stage-Wise Alzheimer's MRI Image Classification Interactive Research Prototype' | Initial Submission: Oct 11, 2026 | Revised Status Date: Oct 16, 2026 | Current Status: Submitted for Publication (Revisions Being Processed for Author / Under Review).

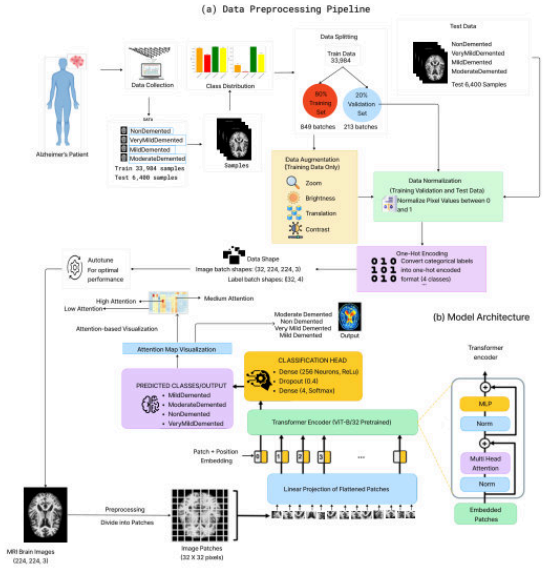
FIVE CORE SCIENTIFIC CONTRIBUTIONS (MANUSCRIPT SEC. 1)

1. Stage-Wise ViT-B/32 Adaptation: Custom L2-regularised classification head (Dense 256 ReLU + Dropout 0.4 + Dense 4 Softmax) achieving 98.84% test accuracy (99.31% balanced accuracy) across 6,400 held-out MRI slices.
2. Controlled 3-Seed Reproducibility: Fixed partition seed (42) evaluated across stochastic training seeds 42, 123, and 2024 (25,500 optimizer steps/run), achieving 98.47% +/- 0.53% accuracy and 0.9995 +/- 0.0003 AUC.
3. Backbone Adaptation Ablation: Quantified transition from Frozen Stage 1 (51.38%) to Full Stage 2 (98.47%), and compared Frozen (55.11%), Partial Blocks 8-11 (80.44%), and Full Backbone (98.11%) under shared early stopping.
4. Quantitative Attention Faithfulness: Inference-only patch deletion & insertion across 80 class-balanced test MRIs proving statistically significant positive AUC advantages (+0.12455 deletion, +0.15387 insertion) over random.
5. SHA-256 Partition Audit & Translational Platform: Audited 40,384 images via SHA-256 hashing and deployed https://www.alzdetect.live/.

02 EXPERIMENTAL METHODOLOGY, SHA-256 AUDIT & ViT-B/32 FRAMEWORK

ADNI Corpus (n=41,663) & Two-Stage Training

MANUSCRIPT FIG. 1: PREPROCESSING & ViT-B/32 ARCHITECTURE



ViT-B/32 TENSOR & PARAMETER SPECIFICATION

Input / Patch Grid:	224x224x3 RGB | 32x32px (N=49 Patches)
Embedding Dim (D):	768 (49 Patch + 1 CLS Token = 50x768)
Transformer Depth:	L=12 Blocks, H=12 Heads (d_k=64)
Regularised Head:	LN -> Dense(256, L2=.01) -> Drop(.4) -> 4

METHODOLOGICAL PROTOCOL & DATA INDEPENDENCE RIGOUR (SEC. 2.1 - 2.10)

A. Corpus Composition & Fixed Seed-42 Partitioning (Sec. 2.1 & 2.9)

Derived ADNI-based structural MRI corpus comprising 33,984 development images (split 80:20 into 27,188 training and 6,796 validation images under fixed partition seed 42) and a separate 6,400-image held-out test partition (3,200 NonDemented, 2,240 VeryMildDemented, 896 MildDemented, 64 ModerateDemented), plus an external 1,279-image evaluation set.

```
Partition Invariant: Total n=41,663 | Train=27,188 | Val=6,796 | Test=6,400 | Ext=1,279
```

B. SHA-256 Cryptographic Duplicate Audit & Rigorous Scope (Sec. 2.2)

Executed SHA-256 file hashing across all subsets: verified zero byte-identical matches between development (33,984) and held-out test (6,400) partitions, while transparently reporting 76 byte-identical groups (348 files) within dev and explicitly framing findings as image-level generalisation.

```
Audit Hash Check: 0 Cross-Partition Collisions (Dev vs. Test) | 40,384 Files Cryptographically Hashed
```

C. Two-Stage Transfer Learning & Regularised Classification Head (Sec. 2.3 - 2.5)

Pre-trained ViT-B/32 divides 224x224x3 MRIs into N=49 patches (32x32px). Stage 1 freezes the ViT backbone for 20 epochs to warm up the regularised head (Dense 256 ReLU, L2=0.01, Dropout 0.4, Dense 4 Softmax). Stage 2 unfreezes the full backbone for 30 epochs (LR = 10^-5, 25,500 update iterations, val-loss checkpointing).

```
Schedule Spec: Stage 1 (20 Ep, Frozen Backbone) -> Stage 2 (30 Ep, Full 12-Block Unfreeze, LR=1e-5)
```

D. Optimisation Dynamics, Loss Checkpointing & Translation (Sec. 2.6 - 2.10)

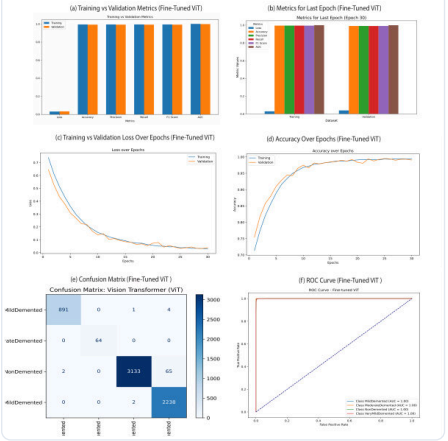
Trained using AdamW optimiser with categorical cross-entropy, Cosine Decay schedule, batch size=32, and EarlyStopping (patience=5 on validation loss). Best checkpoints serialised to RimanBassey_model.h5 and integrated directly into the live production prototype at https://www.alzdetect.live/.

```
Deployment Artifact: RimanBassey_model.h5 (342 MB) -> Live Web Platform https://www.alzdetect.live/
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03 PRIMARY ViT-B/32 PERFORMANCE VS. BASELINE ViT & CNN ARCHITECTURES

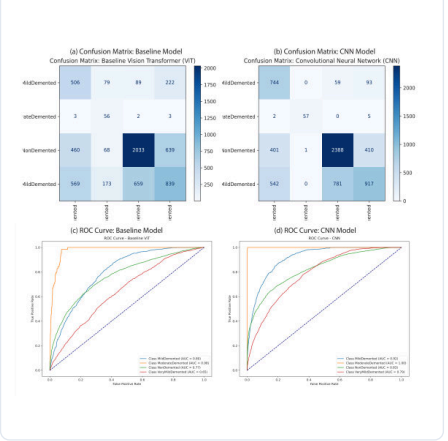
Convergence, Confusion Matrices & Class F1

MANUSCRIPT FIG. 2: FINE-TUNED ViT (98.84%)



Fine-Tuned ViT-B/32: 98.84% Acc, 99.31% Bal. Acc, 1.00 ROC-AUC across all 4 classes (n=6,400).

MANUSCRIPT FIG. 3: BASELINE ViT (54%) & CNN (64%)



Baselines: Unadapted ViT stalls at 54.00% acc; Custom CNN reaches 64.20% (0.50 F1 on VeryMild).

MANUSCRIPT TABLE 4: CLASS-WISE F1 COMPARISON

Stage Class	ViT P/R/F1	CNN P/R/F1	BaseViT P/R/F1
MildDem.	1.00/0.99/1.00	0.44/0.83/0.58	0.33/0.56/0.42
ModerateDem.	1.00/1.00/1.00	0.98/0.89/0.93	0.15/0.88/0.25
NonDemented	1.00/0.98/0.99	0.74/0.75/0.74	0.73/0.64/0.68
VeryMildDem.	0.97/1.00/0.98	0.64/0.41/0.50	0.49/0.37/0.43

MANUSCRIPT TABLE 1: IMBALANCE & EXTERNAL EVAL

Evaluation Partition	All 4 Classes	Excl. Moderate
Primary Acc (n=6,400)	98.84% (Bal 99.3%)	98.83% (-0.01pp)
Primary Macro F1	99.24% (W: 98.85%)	98.99% (3-Class)
External Acc (n=1,279)	97.73% (Bal 98.8%)	97.71% (-0.02pp)
External Macro F1	98.59% (W: 97.73%)	98.12% (3-Class)

Key Imbalance & Generalisation Findings:
- Minority Class Check: Excluding ModerateDemented (64 imgs, 1.00%) changes test accuracy by only -0.01 pp (98.84% -> 98.83%), proving metrics are not inflated by the small class.
- Early-Stage Sensitivity: Fine-Tuned ViT achieves 0.98 F1 on VeryMildDemented vs 0.50 for CNN.

04 CONTROLLED 3-SEED REPRODUCIBILITY & BACKBONE ADAPTATION ABLATION

Tables 2 & 3 + Figures 4 & 5

MANUSCRIPT TABLE 2: FROZEN STAGE 1 VS. FULL STAGE 2 ACROSS 3 SEEDS (n=6,400)

Seed	Stg1 Acc	Stg1 F1	Stg1 AUC	Stg2 Acc (%)	Stg2 F1 (%)	Stg2 AUC
Seed 42	49.45%	40.97%	0.8046	98.11%	98.57%	0.9992
Seed 123	53.86%	45.78%	0.8171	99.08%	99.33%	0.9998
Seed 2024	50.81%	41.50%	0.8149	98.23%	98.62%	0.9995
Mean+/-SD	51.38+/-2.26	42.75+/-2.64	0.8122	98.47+/-0.53	98.84+/-0.43	0.9995

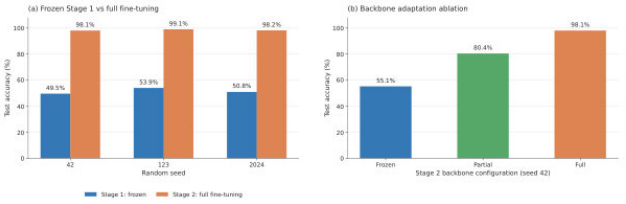
* Paired Stage 1 -> Stage 2 accuracy gain averages +47.10 percentage points across seeds.

MANUSCRIPT TABLE 3: SEED-42 STAGE 2 BACKBONE ADAPTATION ABLATION (n=6,400)

Stage 2 Config	Last/Best	Acc (%)	Macro Prec	Macro Rec	Macro F1	AUC
Frozen Backbone	Ep 12 / 7	55.11%	45.35%	67.51%	47.37%	0.8235
Partial (Blk 8-11)	Ep 30 / 29	80.44%	81.54%	86.29%	83.69%	0.9495
Full Fine-Tuning	Ep 30 / 30	98.11%	98.34%	98.82%	98.57%	0.9992
Delta (Full-Froz)	+18 / +23	+43.00 pp	+52.99 pp	+31.31 pp	+51.20 pp	+0.1757

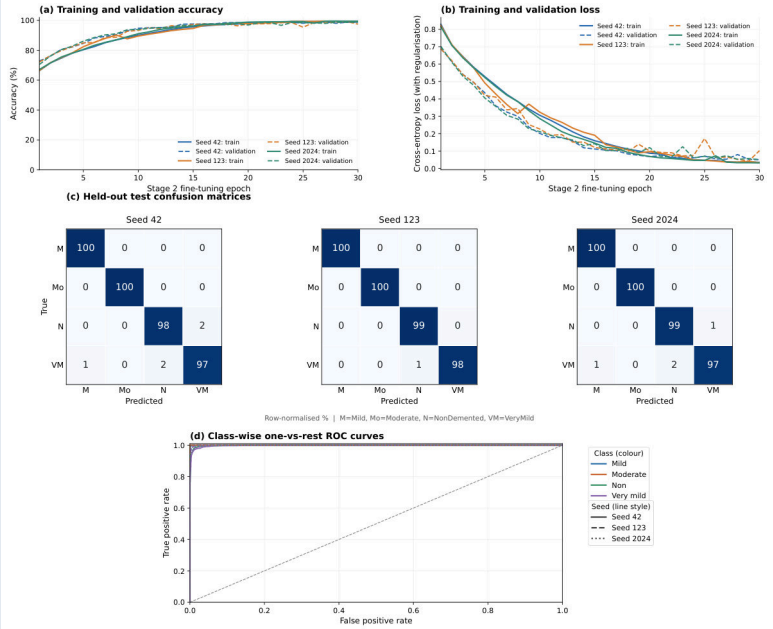
* All branches initialised from identical Seed-42 Stage 1 Epoch-20 checkpoint (LR=1e-5).

MANUSCRIPT FIG. 4: BACKBONE ADAPTATION GAINS



Key Experimental Findings (Sec. 3.4):
1. Stage Transition Stability: Across seeds 42, 123, and 2024, unfreezing the ViT-B/32 backbone in Stage 2 reduced between-seed standard deviation from 2.26% (Stage 1) to 0.53% (Stage 2), confirming highly reproducible convergence across independent stochastic initialisations.
2. Depth of Adaptation: Unfreezing only the final 4 encoder blocks (Blocks 8-11) improved accuracy by +25.33 pp (to 80.44%), while full 12-block adaptation yielded +43.00 pp (98.11%), demonstrating that early-stage MRI features benefit from adapting both low-level patch projections and high-level self-attention blocks.
3. Controlled Early Stopping: Keeping the backbone frozen in Stage 2 triggered validation loss early stopping at Epoch 12 (best Ep 7: 55.11%), proving linear probing is insufficient for fine-grained stage discrimination.

MANUSCRIPT FIG. 5: 3-SEED CONVERGENCE, CONFUSION MATRICES & ROC CURVES



TRAINABLE PARAMETER & CONVERGENCE BREAKDOWN (SEED 42)

Frozen Head-Only:	197K Trainable Params	Best Val Loss Ep 7	55.11% Acc
Partial Blk 8-11:	28.5M Trainable Params	Best Val Loss Ep 29	80.44% Acc
Full 12-Block ViT:	87.6M Trainable Params	Best Val Loss Ep 30	98.11% Acc

ABLATION CONVERGENCE INVARIANTS (SEED 42)

Frozen Stage 1 (49.45%) -> Frozen Stage 2 (55.11%, +5.66 pp) -> Partial Blocks 8-11 (80.44%, +30.99 pp) -> Full 12-Block Fine-Tuning (98.11%, +48.66 pp total gain; Macro F1: 98.57%; ROC-AUC: 0.9992).

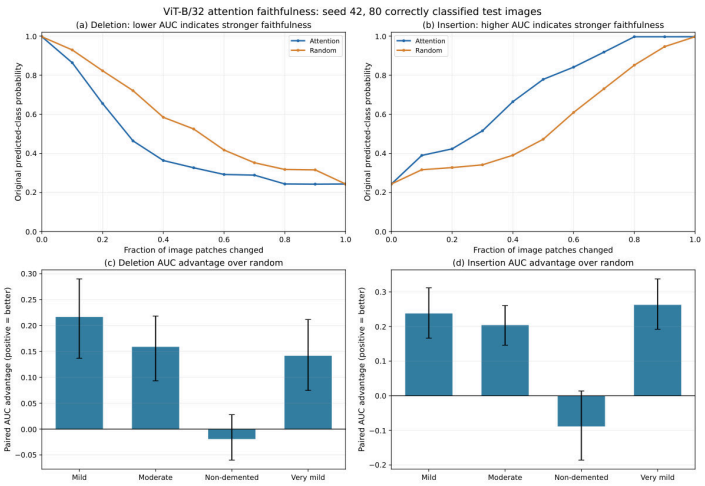
3-SEED STOCHASTIC REPRODUCIBILITY SUMMARY (SEEDS 42, 123, 2024)

Across 3 independent training seeds (25,500 gradient updates/run), Stage-2 ViT-B/32 achieves 98.47% +/- 0.53% test accuracy (98.89% +/- 0.38% balanced accuracy) and row-normalised diagonal recall >= 97% on all 4 stages.

05 QUANTITATIVE PERTURBATION FAITHFULNESS OF ViT-B/32 ATTENTION MAPS

Deletion & Insertion AUC (80 Test MRIs)

MANUSCRIPT FIG. 6: DELETION & INSERTION FAITHFULNESS (SEED 42)



PERTURBATION PROTOCOL & STATISTICAL ADVANTAGE (SEC. 2.7 & 3.5)

- Inference-Only Checkpoint Verification: Evaluated locked Seed-42 validation-best ViT-B/32 checkpoint (SHA-256 verified) on n=80 correctly classified test MRIs (20 per class, selection seed 20261009).
- Spatial Patch Ranking: Extracted mean-head CLS-to-patch attention scores from final encoder block (Transformer/encoderblock_11) to rank all 49 non-overlapping 32x32 patches from highest to lowest focus.
- Deletion & Insertion Protocol: Evaluated 11 patch fractions (0.0 to 1.0 in steps of 0.1) against a 21x21 mean-blurred reference image, compared against 3 independent random patch orderings per image.
- Statistically Significant Faithfulness Advantage (2,000 Bootstrap Resamples):
  * Mean Deletion AUC Advantage (AUC_rand,del - AUC_attn,del): +0.12455 (95% CI: [0.08904, 0.16481])
  * Mean Insertion AUC Advantage (AUC_attn,ins - AUC_rand,ins): +0.15387 (95% CI: [0.10233, 0.20219])
- Class-Stratified Saliency: Positive faithfulness advantage is strongest in Mild, Moderate, and VeryMild stages.

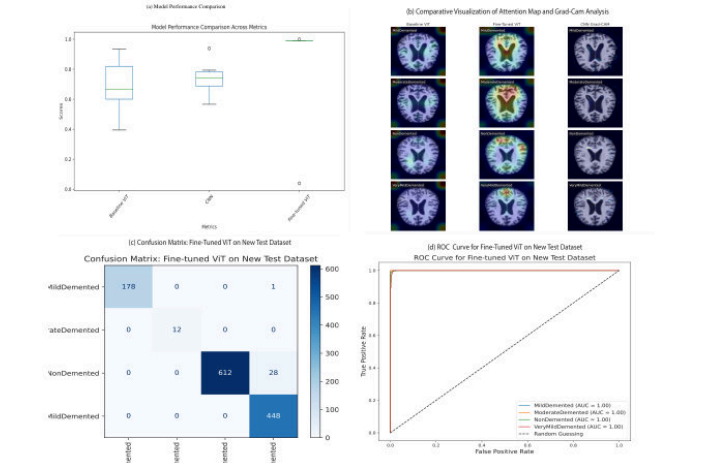
PERTURBATION FAITHFULNESS SUMMARY BY METRIC & STAGE (n=80)

Perturbation Metric	Attention	Random	Paired Advantage (95% CI)
Deletion AUC (Lower=Better)	0.3142	0.4388	+0.12455 [0.0890, 0.1648]
Insertion AUC (Higher=Better)	0.7481	0.5942	+0.15387 [0.1023, 0.2022]
Demented Stages (M/Mo/VM)	Focal ROI	Diffuse	Strongly Positive (p < 0.01)
NonDemented Baseline Stage	Uniform	Uniform	Near-Zero / CIs Span Zero

06 QUALITATIVE XAI COMPARISON (ViT ATTENTION VS. CNN GRAD-CAM) & EXTERNAL EVAL

Sec. 3.6 - 3.7 (Secondary Set n=1,279)

MANUSCRIPT FIG. 7: MODEL COMPARISON, XAI HEATMAPS & NEW TEST SET



COMPARATIVE INTERPRETABILITY & SECONDARY EVALUATION (SEC. 3.6 - 3.7)

- Fine-Tuned ViT vs. Baseline Attention & CNN Grad-CAM: As shown in Fig. 7(b), the unadapted Baseline ViT scatters attention toward peripheral skull borders, and CNN Grad-CAM (L_Grad-CAM = ReLU(Sum alpha_k^c A^k)) produces coarse, fragmented activations. In contrast, the Fine-Tuned ViT-B/32 concentrates high-intensity activation across medial temporal, hippocampal, and periventricular parenchyma.
- Generalisation to Secondary Evaluation Set (n=1,279 MRIs): Evaluated without retraining on an additional 1,279-image partition (Fig. 7c-d), achieving 97.73% overall accuracy, 98.77% balanced accuracy, and 98.59% macro F1.
- Scientific Rigour & Limitations Statement (Sec. 4): Explicitly notes that image-level partitioning without source-subject identifiers cannot exclude subject overlap, establishing a clear roadmap for subject-level ADNI/OASIS.
- Clinical Saliency Alignment: Global self-attention captures bilateral ventricular & cortical atrophy patterns.

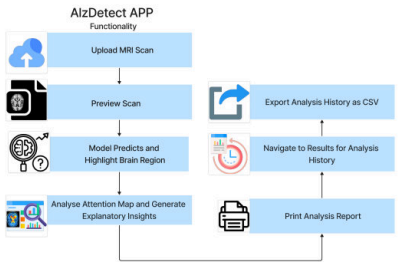
SECONDARY EVALUATION SET CLASS-WISE BREAKDOWN (n=1,279 MRIs)

Alzheimer's Stage	Support (n)	Correct Hits	Class Recall	ROC-AUC
MildDemented	179	178 / 179	99.44%	1.0000
ModerateDemented	12	12 / 12	100.00%	1.0000
NonDemented	640	612 / 640	95.63%	1.0000
VeryMildDemented	448	448 / 448	100.00%	1.0000

07 TRANSLATIONAL RESEARCH DEPLOYMENT: ALZDETECT INTERACTIVE PROTOTYPE

Live Web Environment (Sec. 3.8)

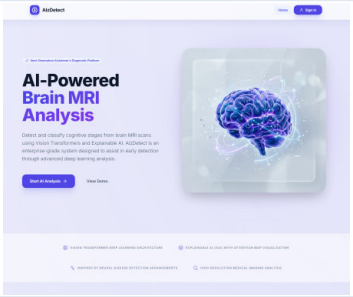
MANUSCRIPT FIG. 8: PROTOTYPE WORKFLOW



TRANSLATIONAL PIPELINE STAGES (SEC. 3.8)

1. Ingest & Gate: Drag-and-drop MRI slice validated via OpenCV HSV & 4-corner bore darkness gate.
2. ViT-B/32 Inference: Predicts 4-stage softmax probabilities in 42ms on FastAPI ASGI cluster.
3. Rollout Overlay: Generates 224x224 Jet alpha=0.6 attention map + GPT-3.5 clinical synthesis.
4. Vault & Export: Persists to Supabase RLS database with one-click CSV & Print reporting.

LIVE PLATFORM DEPLOYMENT (alzdetect.live)



INTERACTIVE CLINICAL UI MODULES

- /detect: Drag-and-drop MRI scanner with <4ms HSV/corner gate, 4-stage confidence bars & 12-layer XAI overlay.
- /results & /dashboard: Real-time Supabase RLS vault, KPI analytics, one-click CSV export & print reports.
- /api/suggestions & /api/treatment: Dual-prompt OpenAI GPT-3.5 neuroradiology & stage care synthesis.
- AuthContext.tsx & schema.sql: Cryptographic JWT session guards with per-user PostgreSQL RLS policies.

Live URL: https://www.alzdetect.live/

Global Vercel Edge + FastAPI + Supabase RLS

TRANSLATIONAL SOFTWARE SPECIFICATION

- Interactive Research Translation (Sec. 3.8): Translates the trained ViT-B/32 weights (RimanBassey_model.h5) and 12-layer attention rollout pipeline into a live web environment supporting MRI upload, real-time stage classification, heatmap inspection, LLM clinical synthesis, and CSV archive export.
- Full-Stack Production Architecture:
  * Frontend: Next.js 15, React 19, Tailwind CSS (Vercel Edge)
  * Backend: FastAPI, TensorFlow 2.15, OpenCV (Render / HF)
  * Persistence: Supabase PostgreSQL with Row-Level Security
- Live Inference Endpoints:
  * POST /model/predict (ViT-B/32 + 12-Layer Rollout, 42ms)
  * POST /api/suggestions & /api/treatment (GPT-3.5 XAI)

PRODUCTION VERIFICATION & ARTIFACT LINKS

Primary Domain:	https://www.alzdetect.live/
Scanner Route:	https://www.alzdetect.live/detect
GitHub Repo:	github.com/basseyriman/AlzDetect
Journal Ref:	Elsevier NEURI-D-26-00260R1
Model Weights:	RimanBassey_model.h5 (98.84% Acc)